

COMP3308 Assignment 2 Part 2: Report

Deadline

Submission: 11:59pm, 15 May 2026 (Friday, week 11)

This part is worth 14 marks = 14% of your final mark

Late submissions policy

Late submissions are allowed for up to 3 days late. A penalty of 5% per day late will apply. Assignments more than 3 days late will not be accepted (i.e. will get 0 marks). The day cut-off time is 11:59pm.

Submission

Each group should submit a pdf file in Canvas. Only one of the students needs to submit, on behalf of the group, not both students.

Report

Describe your analysis and findings in a report which is similar in style to a research paper.

Your report should include the sections listed below. It should be written as if you were describing the study to someone who has not seen the data or this assignment before.

There is no minimum or maximum length for the report. However, please keep your report concise - you will be marked on the quality of the content not the length of the report.

Title and SIDs - e.g.:

COMP3308 Assignment 2 Part 2: Report

Student 1: <SID>

Student 2: <SID>

Student 3: <SID>

Do not include your names as the marking is anonymous.

1. Introduction

This section should briefly state the aim of your study and include a paragraph about why this study is important according to you. This is your interpretation of the task.

2. Data

This section should briefly describe the dataset, mentioning the number and nature of the attributes and classes.

3. The KNN+ and DT+ Algorithms

Describe your KNN+ and DT+ algorithms. How did you modify the standard algorithm and what motivated you to do this? As an Appendix, include the code of your KNN+ and DT+ algorithms with sufficient and helpful comments.

4. Results and discussion

Results

Present the accuracy results in the following tables where MyKNN and MyDT are your implementations of the standard KNN and DT algorithms, and MyKNN+ and MyDT+ are your modified versions of these algorithms, evaluated using your stratified 10-fold cross validation.

The other classifiers in the table are Weka's classifiers. In Weka, use 10-fold cross validation (it is actually 10-fold stratified cross validation) and evaluate the performance of the following algorithms: ZeroR, 1R, k-Nearest Neighbor (kNN; IBk in Weka), Naive Bayes (NB), Decision Tree (DT; J48 in Weka), Multi-Layer Perceptron (MLP), Support Vector Machine (SVM; SMO in Weka), Bagging (Bagg), Boosting (Boost, AdaBoostM1 in Weka) and Random Forest (RF). For Bagging and Boosting, combine decision trees (J48) not the default REPTrees or decision stumps. For the DT classifier, evaluate the algorithm with and without pruning (using the default pruning option in Weka).

Compare the performance of the Weka's classifiers with your K-Nearest Neighbor and Decision Tree classifiers – the core and extended versions.

For the DT classifier in Weka, present two results – for an unpruned tree and for a pruned tree.

For the K-Nearest Neighbor classifiers, present the results for k of your choice.

In addition to accuracy, what other performance measures would be suitable for the task? E.g. recall, precision, F1-measure, training time, others?

Predictive performance – accuracy, other suitable measures

ZeroR	1R	KNN	NB	MLP	SVM	MyKNN	MyKNN+

DT unpruned	DT pruned	MyDT	MyDT+	Bagg	Boost	RF

You may transpose the tables if this makes your results presentation clearer.

DT diagrams

Build DTs using the whole training data and include a DT diagram for your classifiers (MyDT and MyDT+) as an Appendix to the report. Generate a simple text-based diagram as the one produced by Weka.

Discussion

1. Compare the performance of all classifiers in terms of accuracy and other performance measures
2. Compare MyKNN with MyKNN+ and MyDT with MyDT+. Are the results as expected based on your motivation for the modified algorithms?
3. Compare your kNN and DT classifiers with Weka's
4. Compare Weka's DT-unpruned and DT-pruned and discuss the effect of pruning
5. Compare Weka's tree-based classifiers – single tree and ensembles of trees (DT-pruned, Bagging, Boosting, Random Forest)

Include anything else that you consider important.

5. Conclusion

Summarise your main findings and suggest future work.

6. Reflection

Write one or two paragraphs describing the most important thing that you have learned throughout this assignment. Each student should write their own reflection. Include only your SIDs and not your names as the marking is anonymous.

7. Acknowledgement of AI use

This acknowledgement is for both Part 1 (Code) and Part 2 (Report).

You are permitted to use AI tools in this assignments but you will need to include a statement explaining:

- that you have used AI tools in completing the work
- what automated writing or generative AI tools you have used
- how you have used them.

Visit the [Current Students website](#) for more information on AI in assessments, including [details on how to acknowledge its use](#). Additional information about generative AI use is provided on the [AI in Education Canvas site](#). It includes additional guidance on [how to acknowledge and reference the use of generative AI](#).

If you haven't used AI tools, include the following statement: "No AI tools were used in this assignment".

Each student should write their own statement. Include only your SIDs and not your names as the marking is anonymous.